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Intrusion detection system based on hybridizing a modified binary grey wolf optimization and particle swarm optimization

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ABSTRACT

Nowadays, the world is increasingly becoming more connected and dependent on the Internet and Internet-based services. One of the main challenges of interconnectedness is the security of applications and networks from malicious actors. The security challenge is further compounded by the exponential growth of threats and the increase in attack vectors through interfaces of many newly introduced network services. To deal with the security threats, many solutions have been proposed; yet the existing solutions overwhelmingly fail to detect security threats efficiently with high performance. Accordingly, a hybridization of modified binary Grey Wolf Optimization and Particle Swarm Optimization is proposed in this article. The proposed solution uses two benchmarking datasets, NSL KDD'99 and UNSW-NB15, and the results reveal that the proposed solution outperforms the existing solutions, as the proposed approach improves the detection accuracy by approximately 0.3% to 12%, and the detection rate by 2% to 12%. In addition, it reduces false alarm rates by 4% to 43%, and reduces the number of features by approximately 31% to 75%. Last, the proposed approach reduces processing time by approximately 14% to 22% compared to state-of-that-art approaches.

1. Introduction

Intrusion Detection System (IDS) still suffers from one major challenge, which is performance. The performance of IDS can be measured by different metrics, such as detection rate, false negative, accuracy, etc. IDS classifier differentiates normal and abnormal traffic based on the network packet features. Several Machine Learning (ML) algorithms, such as SVM and neural networks, are typically used to perform the classification in the early days. However, the enterprise network is still facing challenging problems due to continuous and rapid changes (Arelion, 2021). Moreover, the introduction of Industry 4.0 expedites the digital transformation process, but at the same time makes cyber-attacks more destructive and far-reaching than ever before (Tseng et al., 2021). Therefore, there is an urgent need for an effective IDS that works with an appropriate set of features that contribute to attack detection.

Moreover, in anomaly detection, many different challenges might degrade training and testing processes' performance and reduce the accuracy and detection rates since the IDS deals with huge volumes of data that comprise irrelevant and redundant features. Therefore, many researchers have proposed solutions based on a statistical approach, rule-based, machine learning, or computational intelligence to address these issues. Computational intelligence-based approaches, such as (Xu et al., 2018; Rathi & Jadhav, 2014; Syarif & Gata, 2017; Resende & Drummond, 2018), are the most efficient approaches that enhance the IDS performance. These approaches select the most significant network traffic features that contribute to detecting certain types of attacks that improve IDS performance. Despite their advantages, these approaches still suffer from selecting many features in the classification process that degrade the IDS performance. The importance of Feature Selection (FS) came from two different perspectives. First, from an IDS perspective, FS is used to remove irrelevant, redundant, and noisy features to enhance IDS performance. Second, from a classification perspective, the FS is

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considered a pre-processing task that minimizes the number of features and error rate of classification (Kumar & Singh, 2016).

Although many researchers have proposed FS based on a single optimization function, such as (Alwan et al., 2020; Syarif et al., 2020), the selected features might be highly redundant and irrelevant, whereas features should have a minimum redundancy and correlation (Alamiedy et al., 2020). The FS field is significant, especially when many features have the same values for many samples. It might not be beneficial for the IDS if the features hold a wide range of values simultaneously. Therefore, a multi-objective function is required to identify the feature subset in a manner that all required properties are preserved to ensure maximum classification performance.

Swarm Intelligence-based algorithms such Grey Wolf Optimizer (GWO) algorithm (Mirjalili et al., 2014) and bat algorithm employed in (Diyaley et al., 2019) have attracted much attention since they can provide possible optimized solutions with high robustness, flexibility, and minimal cost. Further, they can easily solve large-scale sophisticated problems without a need for a centralized control entity. GWO is a metaheuristic inspired by the hunting technique and leadership hierarchy of wolves in nature and has been successfully applied for solving different optimization problems. The GWO shows good performance as a meta-heuristic algorithm for FS problems. However, it provides low precision and slow convergence. A new variant of GWO is used to deal with special problems such as feature selection, the solutions are restricted to the binary values [0,1], which is called binary GWO (bGWO). Moreover, several studies have been conducted to improve the convergence performance of GWO (Tawhid & Ali, 2017; Sharma et al., 2019; El-Kenawy et al., 2020). However, there is still room for improvement in feature selection using GWO by solving optimization problems with multi-objective functions. The features used in this study represent network packet fields found in in the NSL KDD'99 and UNSW-NB15 datasets, such as packet size and duration.

On the other hand, creating a high-performance IDS model is challenging, especially when dealing with network packets with a range of features (i.e., FS problem). The FS is a core stage in the design of IDSs. Since the computational cost and processing time can be reduced by using optimized feature sets, the classification accuracy could be increased, and the false alarm rate could be reduced. Several works have been proposed in the literature to address the FS problem (i.e., selecting a large number of features) (Shakya & Makwana, 2017; Aljawarneh, Aldwairi, & Yassein, 2018; Kamarudin et al., 2019; Pham et al., 2018; Sasan & Sharma, 2016). However, these approaches suffer from inappropriate FS due to inefficient fitness function, which evaluates the selected features' significance.

An integration between classification enhanced computational intelligence-based FS approach, and fitness function could improve the IDS to detect attacks with higher accuracy and detection rate, and minimum false alarm rate. Therefore, this article proposes an efficient IDS that ensures optimal results in terms of the accuracy, number of selected features, detection rate, and false alarm rate.

1.1. Article contributions

This article contributes to the body of knowledge by proposing an efficient IDS for detecting attacks against computer networks by hybridizing feature selection optimization techniques. Other contributions of the article are as follows:

(i) An enhanced bGWO by proposing a modified change position technique,
(ii) Hybridized modified bGWO and binary Particle Swarm Optimizer (bPSO) to enhance the performance of IDS, and
(iii) A multi-objective fitness function to enhance the performance of feature selection and classification process by selecting the most important features.

1.2. Article organization

This article is organized as follows: Section 2 provides synthetic analysis of related works. Section 3 presents the proposed IDS and describes its stage in detail. Section 4 illustrates the experimental settings and discusses the results. Finally, Section 5 concludes this article and provides future work.

2. Related works

Roopa Devi and Suganthe (2020) proposed a hybrid IDS using filter-based methods and wrapper methods. First, the filter-based principle is implemented to minimize redundancy via mutual information, and then the wrapper approach guides the classifier performance. However, the false-positive rate is dismissed. Besides, Elngar et al. (2013) tried to address the issue pertaining to the high feature number for intrusion detection by incorporating the PSO and information entropy minimization method with an unknown naive base classifier. The proposed technique and its performance are assessed using the thins-KDD data set, which yielded a minimized number of features despite no improvement for the accuracy rate.

Also, Shurman et al. (2019) combined the signature-based technique and anomaly-based technique to produce a hybrid IDS technique, which solves most of the existing literature problems such as weak protections against any new attacks and the need to update on regular bases before securing a network. The findings observed that the signature-based uses a database with fast detection time, but these methods do not have the detection capabilities for any new attack types. On the other hand, anomaly-based methods do not use any database to recall attacks log history but can detect new attack types and has varying computation time; besides, both have reliable results. However, the proposed hybrid method depends on retrieving data from a database and has reliable results; it can detect suspicious attacks with varying detection time.

Khan et al. (2019) proposed a hybrid IDS based on the convolutional-LSTM network and Spark ML to overcome the existing IDS issues. One of the issues is detecting new security threats due to reliance on classical machine learning methods that gives less focus on FS and classification accuracy. Therefore, a lot of known signatures from the attack traffic stay unknown and become latent. The proposed IDS overcomes these issues and enhance the accuracy and scalability. It contains two-main stages, namely: (i) the anomaly detection module using Spark ML, and (ii) the misuse detection module using Conv-LSTM network. Results revealed that the proposed IDS could identify network misuses in 97.29% of cases and outperforms existing approaches using the ISCX-UNB dataset.

Venkatraman & Surendiran (2020) proposed a hybrid IDS based on signature and anomaly IDS. The proposed IDS aims to adapt to dynamic service changes in real-time by using a timed automata controller. It has extra knowledge using multimedia file formats. Results reveal that the proposed IDS achieved 99.06% detection accuracy in IoT environments. Given that detection effectiveness is the most crucial security concern, the proposed IDS should provide high effectiveness. Each existing IDS technique's effectiveness level is evaluated by demonstrating accuracy, detection rate, false alarm, processing time, and number of features used (Alzoubi et al., 2020). Most existing IDS improve some effectiveness measures and ignore others. For instance, Alamiedy et al. (2020) enhances detection accuracy and ignores other essential effectiveness measures, such as the number of features that negatively affects detection effectiveness. Other studies such as (Velliangiri, 2020) ignore the processing time of attack detection. On the other hand, some existing IDS techniques (Devi & Suganthe, 2017) were not considered the type of attacks they detect. Since every attack type has its mechanism and it should be detected efficiently to minimize its effect on the network.

As aforementioned, feature selection is a critical issue in classification tasks. By selecting a proper feature subset, the reduction of dataset dimensionality is achieved; therefore, decreasing the classification

computation complexity and increasing the classification performance by avoiding irrelevant or duplicate features. Most existing techniques treat the FS as a single-objective optimisation problem. However, using multi-objective approaches to the FS problem improves the classification performance and provides the generalisation in the supervised classifiers case, or counterbalance the bias toward the number of features that present some techniques utilized to validate the classification in unsupervised classifiers case. Several swarm intelligent algorithms have been proposed for FS, but GWO and its variants, such as bGWO, have many advantages over other algorithms, such as easier implementation due to its simple structure, lower storage, and computational demands, faster convergence due to continuous reduction of search space, fewer decision variables (i.e., α, β, and δ), they can avoid local minima, and only have two control parameters (i.e., a and C) for algorithm tuning resulting in stable and robust performance. Additionally, GWO has been implemented in different fields, such as scheduling to solve the optimization problem with impressive results (Liu et al., 2021). Therefore, bGWO algorithm has been selected among other swarm intelligent algorithms in this study.

Based on the list of computer network security issues, there is a need for an efficient IDS technique for securing the network to be proposed. This technique should be able to overcome the key issues of the existing IDS techniques, including the detection of attacks with high accuracy, high detection rate, low false alarm rate, fewer features, and less processing time.

3. Proposed IDS

The feature selection techniques are commonly used in IDS for efficient attack detection and classification with high performance in a timely and automatic manner. However, many challenges remain since the attack types keep changing, especially involving a huge volume of traffic data, therefore requiring a scalable and efficient IDS solution for efficient attack detection, which this article is proposing. As shown in Fig. 1, the proposed IDS consists of three main stages, namely: (i) Data Preparation, which pre-processes the datasets. The datasets are pre-processed and normalized to give appropriate format to be used as input to the next stages. (ii) Feature Selection, which is the main stage in the proposed technique, eliminating redundant, irrelevant, and noisy features to reduce the dataset's dimensionality to improve the classifier's performance. This stage includes two main sub-stages: (a) enhancing bGWO by modifying the mechanism for identifying the next position and using a multi-objective function as a fitness function, (b) hybridizing modified bGWO and bPSO to select the most relevant features. (iii) Classification stage, which adapts the SVM and decision tree classifiers to evaluate the selected features from stage 2, which detects abnormal activities (attacks) through the SVM and decision tree classifiers.

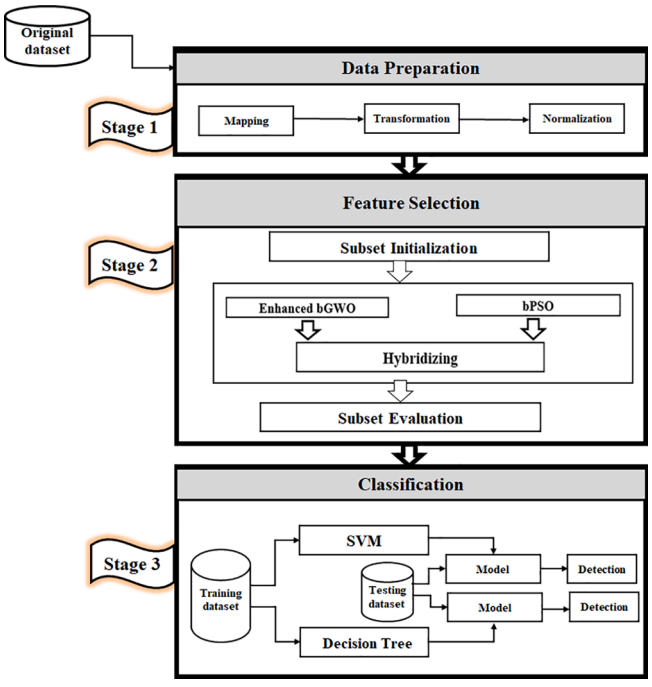


Fig. 1. Architecture of bGWbPS-IDS.

3.1. Data preparation

The aim of Stage 1 (data preparation) is to prepare the data for the subsequent stages of bGWbPS-IDS (features selection and classification stages) by transforming the gathered network traffic into a more meaningful form. Generally, every dataset contains a lot of meaningless information, including irrelevant and redundant data that might increase the computational time and degrades the detection performance (Romero et al., 2014). Data preparation is a vital and common stage in many different IDS techniques, which removes inaccurate or irrelevant data to avoid high false alarm results. As shown in Fig. 2, this stage consists of three main steps: (i) mapping, (ii) transformation, and (ii) normalization.

In detail, each attack is mapped into its attack category in the data preparation stage. Fig. 3 shows the attacks' mapping process. Consequently, all abnormal dataset's traffic should belong to one of the attack categories (i.e., DoS, Probe, R2L, or U2R). Manual mapping using MS-Excel is used to map each attack into its corresponding attack category, for instance instances of Smurf and Neptue are mapped into DoS, Satan and Ipsweep are mapped into Probe, Warezlient and Guess_password are mapped into R2L, and Buffer_overflow and Rootkit are mapped into U2R.

Then, the features of network traffic that contains symbolic values (i.e., protocol type, service value, flag value, classification of attack) are transformed to numerical values as shown in Table 1 below. (See Appendix A for details of the transformation conducted on UNSW-NB15 dataset).

Data transformation followed by normalization (refer to Fig. 2), which is applied to the dataset to ensure all values are binary (i.e., range only belongs to [0, 1]) to be used for extracting the features in the next stage of bGWbPS-IDS. Fig. 4 shows a snapshot of the dataset after the normalization and mapping process.

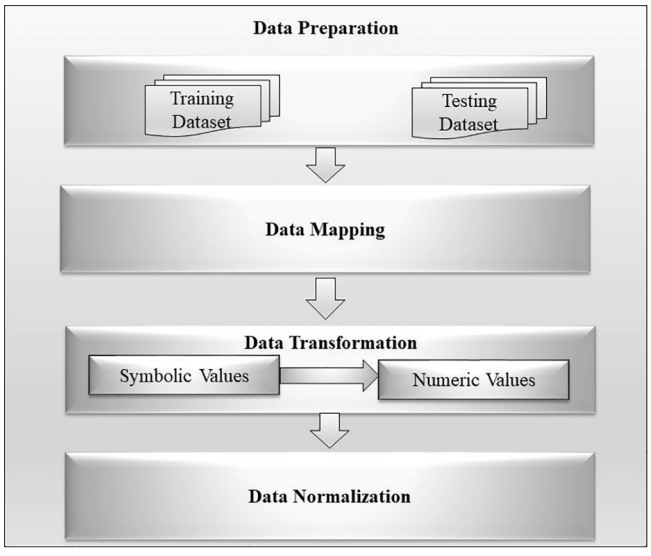


Fig. 2. The steps of Stage 1 (Data Preparation).

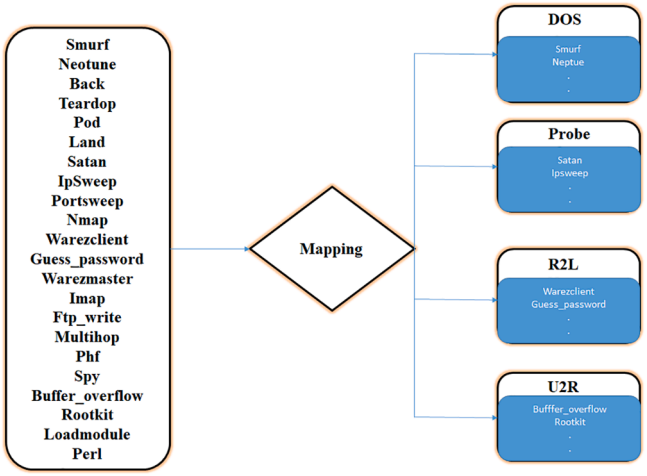


Fig. 3. Attacks Mapping.

Table 1
Transformation of symbolic values to numerical values for NSL KDD'99.

Feature	Values
Protocol type	tcp = 1, udp = 2, icmp = 3
Service value	private = 1, ftp_data = 2, eco_i = 3, telnet = 4, http = 5, smtp-6, ftp = 7, Idap = 8, pop_3 = 9, courier = 10, discard = 11, ecr_i = 12, imap4 = 13, domain_u = 14, mtp = 15, systat = 16, iso_tsap = 17, other = 18, csent_ns = 19, finger = 20, uucp = 21, whois = 22, netbios_ns = 23, link = 24, Z39_50 = 25, sunpc = 26, auth = 27, netbios_dgm = 28, uucp_path = 29, vmnet = 30, domain = 31, name = 32, pop_2 = 33, http_443 = 34, urp_i = 35, login = 36, gopher = 37, exec = 38, time = 39, remote_job = 40, ssh = 41, kshell = 42, sql_net = 43, shell = 44, hostnames = 45, echo = 46, daytime = 47, pm_dump = 48, IRC = 49, Netstat = 50, ctf-51, nntp = 52, netbios_ssn = 53, tim_i = 54, supdup = 55, bgp = 56, nnsp = 57, rje = 58, printer = 59, efs = 60, X11 = 61, ntp_u = 62, klogin = 63, tftp_u = 64, red_i = 65, urh_i = 66, http_8001 = 67, aol = 68, http_2784 = 69, harvest = 70
Flag	REJ = 1, SF = 2, RSTO = 3, S0 = 4, RSTR = 5, SH = 6, S3 = 7, S2 = 8, S1 = 9, RSTOS0 = 10, OTH = 11
Attack class	Dos = 1, Probe = 2, R2L = 3, U2R = 4

3.2. Feature selection

Feature selection process ensures the selection of the optimum subset of relevant features that represents an original feature set with lowest error rates in a classification model design. It is considered a significant pre-processing stage in IDS since it reduces the data dimensionality and enhances the classification process performance. The FS stage is divided into two sub-stages, as follows:

3.2.1. Modified bGWO

The involvement of more wolves in bGWO will lead to a more accurate decision and minimize decision dependency. The modified bGWO is based on this concept by adding one more wolf, called omega wolf (ω). However, the existing bGWO IDSs used a single objective function, which leads to inappropriate selection of features; therefore, there is a need to propose a multi-objective fitness function.

The initial population generation is the first step in the functioning of a modified bGWO. It is the process of generating an initial population (position of features). The initial population is generated randomly so that each member of this population encodes a possible solution to a problem. Then the initial set of features resulted from the modified bGWO is assigned to the classifier and fitness function to determine if the selected feature set is good or not. Therefore, the algorithm has a better possibility of finding a good solution. However, if the initial population size is not big enough, it would be difficult for the algorithm to find an optimal solution. The initial population generation follows Eq. (1).

$$pf = rand(nw, nf)*(ub + lb) + lb \tag{1}$$

where pf is the position of subset features for classification, nw is the number of agents (i.e., grey wolves) in the solution space, nf is the number of the features in the data set, ub is the max value for feature, and lb is the minimum value of the feature.

This is a core stage of this research, and it aims to enhance the bGWO. The enhancement consists of two main parts: proposed fitness function and enhanced wolves' position mechanism.

- **Proposed fitness function**

The proposed fitness function is the basic component of the modified bGWO. It aims to evaluate if a feature subset meets the objectives (i.e., high detection accuracy and minimum number of features). The proposed fitness function is a multi-objective function, which aims to select a small subset of features with high detection accuracy.

The majority of existing IDS depends on single objective function as a fitness function (either simple or complex single objective function), which result in the number of features used remain large; this poses extra network overhead and requires more computation time. Therefore, a multi-objective function is used as a fitness function in the

0.00000	0.33	0.03	0.18	0.0000	0.0000	0	0.0	0.0	0.00	0.00	0	0.0000	0	0	0.0000	0.00	0.0	0.000	0	0	0	0.00391	0.00391	0.00
0.00000	0.67	0.27	0.18	0.0000	0.0000	0	0.0	0.0	0.00	0.00	0	0.0000	0	0	0.0000	0.00	0.0	0.000	0	0	0	0.02544	0.00196	0.00
0.00000	0.33	0.01	0.36	0.0000	0.0000	0	0.0	0.0	0.00	0.00	0	0.0000	0	0	0.0000	0.00	0.0	0.000	0	0	0	0.24070	0.01174	1.00
0.00000	0.33	0.07	0.18	0.0000	0.0016	0	0.0	0.0	0.00	0.00	1	0.0000	0	0	0.0000	0.00	0.0	0.000	0	0	0	0.00978	0.00978	0.20
0.00000	0.33	0.07	0.18	0.0000	0.0001	0	0.0	0.0	0.00	0.00	1	0.0000	0	0	0.0000	0.00	0.0	0.000	0	0	0	0.05871	0.06262	0.00
0.00000	0.33	0.01	0.09	0.0000	0.0000	0	0.0	0.0	0.00	0.00	0	0.0000	0	0	0.0000	0.00	0.0	0.000	0	0	0	0.23679	0.03718	0.00
0.00000	0.33	0.01	0.36	0.0000	0.0000	0	0.0	0.0	0.00	0.00	0	0.0000	0	0	0.0000	0.00	0.0	0.000	0	0	0	0.32485	0.01761	1.00
0.00000	0.33	0.01	0.36	0.0000	0.0000	0	0.0	0.0	0.00	0.00	0	0.0000	0	0	0.0000	0.00	0.0	0.000	0	0	0	0.22896	0.03131	1.00
0.00000	0.33	0.60	0.36	0.0000	0.0000	0	0.0	0.0	0.00	0.00	0	0.0000	0	0	0.0000	0.00	0.0	0.000	0	0	0	0.52838	0.04501	1.00
0.00000	0.33	0.01	0.36	0.0000	0.0000	0	0.0	0.0	0.00	0.00	0	0.0000	0	0	0.0000	0.00	0.0	0.000	0	0	0	0.26027	0.01566	1.00
0.00000	0.33	0.01	0.09	0.0000	0.0000	0	0.0	0.0	0.00	0.00	0	0.0000	0	0	0.0000	0.00	0.0	0.000	0	0	0	0.40117	0.02348	0.00
0.00000	0.33	0.01	0.36	0.0000	0.0000	0	0.0	0.0	0.00	0.00	0	0.0000	0	0	0.0000	0.00	0.0	0.000	0	0	0	0.38943	0.00587	1.00
0.00000	0.33	0.07	0.18	0.0000	0.0004	0	0.0	0.0	0.00	0.00	1	0.0000	0	0	0.0000	0.00	0.0	0.000	0	0	0	0.00587	0.01370	0.00
0.00000	0.33	0.03	0.18	0.0000	0.0000	0	0.0	0.0	0.00	0.00	1	0.0000	0	0	0.0000	0.00	0.0	0.000	0	0	0	0.00391	0.00391	0.00
0.00000	0.33	0.48	0.36	0.0000	0.0000	0	0.0	0.0	0.00	0.00	0	0.0000	0	0	0.0000	0.00	0.0	0.000	0	0	0	0.45597	0.00196	1.00
0.00000	0.33	0.34	0.36	0.0000	0.0000	0	0.0	0.0	0.00	0.00	0	0.0000	0	0	0.0000	0.00	0.0	0.000	0	0	0	0.18787	0.03131	1.00
0.00000	0.33	0.07	0.18	0.0000	0.0027	0	0.0	0.0	0.00	0.00	1	0.0000	0	0	0.0000	0.00	0.0	0.000	0	0	0	0.01566	0.01761	0.00
0.00000	1.00	0.04	0.18	0.0000	0.0000	0	0.0	0.0	0.00	0.00	0	0.0000	0	0	0.0000	0.00	0.0	0.000	0	0	0	0.00196	0.00196	0.00
0.00000	0.33	0.07	0.18	0.0000	0.0001	0	0.0	0.0	0.00	0.00	1	0.0000	0	0	0.0000	0.00	0.0	0.000	0	0	0	0.00587	0.00587	0.00
0.00000	0.33	0.07	0.18	0.0000	0.0002	0	0.0	0.0	0.00	0.00	1	0.0000	0	0	0.0000	0.00	0.0	0.000	0	0	0	0.01761	0.01957	0.00
0.00000	0.33	0.22	0.36	0.0000	0.0000	0	0.0	0.0	0.00	0.00	0	0.0000	0	0	0.0000	0.00	0.0	0.000	0	0	0	0.43640	0.04501	1.00
0.00000	0.33	0.01	0.36	0.0000	0.0000	0	0.0	0.0	0.00	0.00	0	0.0000	0	0	0.0000	0.00	0.0	0.000	0	0	0	0.54795	0.03327	1.00
0.00000	0.33	0.07	0.18	0.0000	0.0023	0	0.0	0.0	0.00	0.00	1	0.0000	0	0	0.0000	0.00	0.0	0.000	0	0	0	0.01566	0.01957	0.00

Fig. 4. Snapshot of dataset after preparation stage.

proposed IDS to overcome the challenges that the existing IDSs are still suffering form.

The existing multi-objective functions used for IDS relies on assigning a weight (w) to each objective function f(x). Therefore, the problem is converted to a single objective with a scalar objective function as follows:

$$Maxi = w1.i^{'}(x1) + \cdots.. + wn.i^{'}(xn) \tag{2}$$

where i'(x) is the normalized objective function, and wn is the weight for objective function (n). The summation of all weights must be 1.

This approach is called the "weighted sum approach" (Kalita et al., 2019). The weighted sum approach is considered an optimal multi-objective optimization approach, as its implementation is straightforward. However, this approach's main difficulty is in selecting an optimal weight vector for each objective. In this article, the following equation is used to compute the weights (Gunantara, 2018):

$$w_x = \frac{2(t+1-x)}{t(t+1)} \tag{3}$$

where, Wx denotes weight for objective (x) (i.e., accuracy and number of features), t denotes total number of objectives, and i denotes the rank number of weight (Wx) in the main formula.

- **An enhanced wolves position mechanism**

As aforementioned, adding one more wolf to the existing three wolves might provide a more accurate decision and minimize decision dependency. The modified bGWO is based on this idea by adding an omega wolf (ω).

This section aims to explain the enhancement of the next position of bGWO. The bGWO was unrestricted to GWO for the feature selection problem. However, in a modified bGWO, the next position changed based on the four best solutions of α, β, δ, and ω using the crossover approach. The bGWO is enhanced by adding the omega wolf to participate in changing the grey wolves' positions.

The increase in the number of wolves that participated in the decision led to a reduction in the impact rate of any wolf's decision from 0.33% to 0.25%. This is described in detail in Eqs. (4)–(8). The central updating equation is formed in Eq. (4).

$$W_i^{t+1} = "Crossover"(w_1.w_2.w_3.w_4) \tag{4}$$

where w1, w2, w3, and w4 are binary vectors that represent the wolf move impact on alpha, beta, delta, and omega grey wolves in sequence. w1, w2, w3, and w4 were mathematically determined based on Eq. (8).

$$w_w^d = \begin{matrix}\{1\&if \quad (w_\omega^d + \ stepb_\omega^d) \geqslant 1 \\ 0\&otherwise\}\end{matrix} \tag{5}$$

where w_ω^d is the location vector of the omega wolf in d, and $stepb_\omega^d$ is a binary step in dimension d which determined by Eq. (6).

$$stepb_\omega^d = \begin{matrix}\{\ 1\&\ if \quad stepc_\omega^d \geqslant rand \\ 0\&otherwise\}\end{matrix} \tag{6}$$

where rand is a randomly selected number from uniform distribution $\in [0.1]$ and $stepc_\omega^d$ is the continuous valued step size for dimension d. Eq. (7) is used to calculate the sigmoidal function.

$$stepc_\omega^d = \frac{1}{1+e^{-10(A_4^d D_{i\omega}^d - 0.5)}} \tag{7}$$

where A_4^d, and $D_{i\omega}^d$ were mathematically determined in dimension d.

A simple strategy of random probability distribution crossover was implemented per dimension to crossover w1, w2, w3, and w4 outcomes, as illustrated in Equation (8).

$$w_d = \{w_1^d if \ \ rand < \frac{1}{4}$$

$$w_2^d \frac{1}{4} \leq rand < \frac{2}{4}$$

$$w_3^d \frac{2}{4} \leq rand < \frac{3}{4}$$

$$w_4^d \ \ otherwise\} \tag{8}$$

The modified bGWO can be considered as an enhanced version of the original bGWO in (Emary et al., 2016). However, it displayed varying solution update (next position move) process based on positions of four wolves, instead of three. The modified bGWO also exerted different decision impact rate calculation, which exhibited a great impact on the algorithm's search performance. After finishing the update of the next location, the termination criteria are checked to stop the FS process. There are two conditions to stop the FS process; either the selected sub features' fitness values obtain the specific values of accuracy, or the number of iterations exceeds the maximum number of iterations. The output of this stage achieves the first and third research objectives. Fig. 5 summarizes the logical steps of modified bGWO.

In fact, adding one more wolve provides a more accurate decision and minimize decision dependency; and therefore, the overall performance, especially in terms of accuracy, false alarm, and number of features, is enhanced. Although, the experiments revealed that there are trivial increases (i.e., not significant) in processing time might be occur due to this improvement, it provides significant improvement in performance, which is a major concern in IDS.

3.2.2. Hybridizing modified bGWO with bPSO

In the bPSO, the particles depend on two values: Gbest and Pbest to drive the search to reach the global optima. At each iteration, the bPSO evolves the swarm based on selecting some values attracted by Gbest and Pbest solution. By means of this operation, the convergence speed of bPSO can reach the stagnation situation very quickly. Therefore, the bPSO can converge prematurely and be trapped into a local minimum, especially with complex problems involving a large feature set. The reason for premature convergence is because the swarm will be identical since the Gbest and Pbest solutions are the only source of the evolution

Initialize the grey wolf population Wi(i=1,2,……n,
Initialize b, A, and C
Calculate the fitness of each search agent
wα = the first best search agent
wβ = the second best search agent
wδ = the third best search agent
wω =the fourth best search agent
While (iter < Max number of iterations)
For each search agent
Calculate w1, w2, w3, and w4
Calculate wd
End for
Update b, A, and C
Calculate the fitness of all search agent
Update wα, wβ, wδ, and wω
iter=iter+1
end while
return wα

Fig. 5. Modified bGWO.

process, making the bPSO lose diversity (Han & Liu, 2014; Jiang et al., 2019). In order to avoid such problem, an enhanced version of bPSO is proposed by hybridizing a modified bGWO concept with bPSO (abbreviated as bGWbPS), as follows:

- In the step to generate the initial population and parameters, the bPSO procedure starts with random initialization of the generated initial particle swarms and its control parameters. Consequently, the initial particle positions are determined by Eq. (1). The velocity of any particle is presented as a vector from its former position to the current position, while the initial velocity of all particles is set to zero. In this step, no changes are made against the original modified bGWO.
- In the step to calculate the fitness values, initial swarm is generated followed by calculating the fitness values for all particles using Eq. (9). In spot, fitness function is calculated for each generated swarm in enhanced bGWbPS while retaining the four best particles as done for the modified bGWO. In the original bPSO, the Gbest and Pbest are retained.
- In the step to update the particles velocity, the velocity of each particle in the original bPSO algorithm is affected by the particle's positions with the best fitness (Gbest) as well as each particle's own best fitness (Pbest). While in the proposed enhanced bGWbPS, the particles are evolved using the velocity in Eq. (8).
- Fitness refers to how well a particle performs. In a flock of birds, the fitness might represent a bird's closeness to a food source. In an optimization algorithm, fitness might refer to the proximity of a particle to optima. It is important to note that the process involves social interaction and intelligence so that the birds learn from their own experience (local search) and the experience of others around them (global search).
- A premature convergence between using local search (Pbest) and global search (Gbest) is a problem that usually occurs when using the standard bPSO. In other words, once the particles in the standard bPSO tend to move closer to Gbest that is a local optimum, they will lose diversity.
- To cope with the diversity issue, the hybridization mechanism of the proposed bGWbPS keeps changing particles velocity based on four best solutions instead of two, which are called Gbest (α), Pbest (β), third-best (δ), and fourth-best (ω). This is inspired by the idea of modified bGWO where the four best solutions are used to manage the search process.
- Each particle updates its velocity Vi to catch up with the four best particles. This change is applied in Equation and uses this new velocity to update its position Xi as represented by Eq. (9).

$$V_i(t+1) = w*V_i(t) + c_1*r_1*(\alpha - X_i(t)) + c_2*r_2*(\beta - X_i(t)) + c_1*r_1*(\delta - X_i(t)) + c_2*r_2*(\omega - X_i(t)) \quad (9)$$

where c1 and c2 are two positive constants called acceleration constants, r is a random function that generates a uniform value in the range of [0,1], and w is an inertia weight, Vi(t) denotes the velocity of the particle (t), and Xi(t) denotes the position of the particle (t). The particle's position and velocity are updated at each time step until a termination condition is reached. The search process is stopped automatically once the predetermined output, or the maximum number of iterations is achieved. The selected features resulting from this stage are used as an input for the next stage (refer to Fig. 6 for more explanation).

The selected features resulting from this stage are used as input for the next stage. The output of this stage achieves the second research objective.

In modified bGWO, the wolves keep changing their position based on four best solutions, α, β, δ, and ω to reduce the impact of the best solutions. The bGWO used the crossover approach to change the position of the grey wolves to a point in the space based on three best solutions; α, β, and δ in GWO to perform FS. Fig. 7 illustrates the proposed modified bGWO FS algorithm.

Generate initial population randomly
Generate initial parameters
Calculate the fitness of each particle swarm

Determine the first four best particles based on fitness

Update the velocity of the current particle

Update the position of the current particles

IF termination criteria is met or t> max

Introduce best solution
End
Go to fitness calculation step

Fig. 6. Hybridizing bGWO with bPSO.

As aforementioned, the bGWO was unrestricted to GWO for FS problem. However, in improved bGWO, the next position changed based on the four best solutions of α, β, δ, and ω using the crossover approach. It is the motivation behind adding the omega wolf in the special version of GWO to participate in changing the positions of the grey wolves via Eq. (4). The increase in the number of wolves that participated in the decision led to reduction in the impact rate of the decision of any wolf from 0.33 to 0.25.

The modified bGWO can be considered as an enhanced version of the original bGWO. However, it displayed varying solution update (next position move) process based on the four wolves' positions, instead of three. The improved bGWO also exerted different decision impact rate calculations, which significantly impacted the algorithm's search performance. Finally, although the new position update process of the modified bGWO algorithm increased the algorithm's overall processing time, it improved the overall performance of the algorithm.

Initialize the grey wolf population Wi(i=1,2,.....n)
Initialize b, A, and C
Calculate the fitness of each search agent
w_α = the first best search agent
w_β = the second best search agent
w_δ = the third best search agent
w_ω =the fourth best search agent
While (iter < Max number of iterations)
  For each search agent
    Calculate w1, w2, w3
    Calculate wd
  End for
  Update b, A, and C
  Calculate the fitness of all search agent
  Update w_α, w_β, w_δ, and w_ω
iter=iter+1
end while
return w_α

Fig. 7. Modified bGWO Algorithm.

However, the original bPSO can converge prematurely and be trapped into a local minimum due to the diversity loss situation. However, the performance of bPSO can be further enhanced by utilizing some important concepts of a modified bGWO to enrich the exploration process. The proposed method is called an enhanced bGWbPS. Instead of using only the Gbest and Pbest in bPSO, the best four solutions selected using the modified bGWO are utilized. In the improvement loop of bPSO, as modified in Eq. (9). Any generated swarm will be affected by the best four solutions in the population. Through this modification, the enhanced bGWbPS enabled diversifying the search around four solutions instead of only two. The four best solutions are iteratively updated using Eq. (9). Therefore, the diversity of enhanced bGWbPS is appropriately controlled; thus, the search will remain effective until the last course run.

3.3. Classification

This stage uses the SVM and decision tree classifiers to evaluate the selected features resulting from Stage 2 to detect abnormal activities (attacks). The classifiers are used as prediction models based on the selected features from the previous stage (i.e., the feature selection stage). The selection of these machine learning classifiers is based on an extensive review of the existing machine learning method used in IDS. The review reveals that the SVM and decision tree are commonly used in IDS such as (Al-Yaseen et al., 2017; Zou et al., 2018; Lin et al., 2012) with relatively high efficiency and effectiveness (Amarudin, 2020). The proposed hybrid method is used to select the most significant features that contribute to detecting malicious attacks. Then, the selected features will be used to train the model using SVM and decision tree classifiers. The training model will be operating online to detect malicious attacks. The online mode of the classifiers has been simulated using supplied testing set approach, as shown in Tables 5 and 7.

4. Experiments and analysis

This section provides the analysis and evaluation of the proposed IDS in terms of detection accuracy, detection rate, and processing time using two different benchmarking datasets (NSL KDD'99 and UNSW-NB15).

4.1. Experimental setups

This section explains the setup configuration, benchmarking datasets, evaluation metrics, and hardware and software specifications.

4.1.1. Configuration setups

Since the proposed IDS is based on different algorithms, the parameters must be properly configured to provide an optimal solution. The value of each parameter related to bGWO and bPSO is adopted from several existing studies (Al-Tashi et al., 2019; Sundaramurthy & Jayavel, 2020). Similarly, for the SVM and decision tree, the value of each parameter is adopted from several existing studies (Ahmad et al., 2019; Ahmim et al., 2019; Gauthama Raman et al., 2019; Al Shorman et al., 2020). These parameters' values are adopted from existing studies to ensure a fair comparison. Meanwhile, the optimal parameter values for the number of wolves, number of iterations, initial weight, and weighting factor, are determined via intensive experiments for selection in the setup configuration. Table 2 presents the configuration parameters.

4.1.2. Benchmarking dataset

Two main benchmark datasets were used to evaluate the performance and robustness of the proposed bGWbPS-IDS in terms of detection accuracy, the number of features, and processing time. These datasets are NSL KDD'99 and UNSW-NB15. These datasets are used to cover different types of attacks and to involve different network packet's features. Although the NSL KDD'99 has been used to evaluate the effectiveness of many IDSs, it still suffers from multiple issues. For example, a lack of recent low footprint attack types, lack of updated normal network traffic scenarios, and a different distribution of training sets and testing sets (Moustafa & Slay, 2016), therefore, UNSW-NB15 is also used to evaluate the proposed bGWbPS-IDS. Table 3 shows the statistics of benchmarking datasets.

Table 2
Configuration Parameters.

Algorithm	Parameters	Symbol	Value
bGWO bPSO	Number of wolves	Nagents	12
	Number of iterations	NIter	20
	Initial weight	wMax	0.9
	Final weight	wMin	0.4
	Weighting factor	C	0.5
	Uniformly distributed random number	Rand ()	0–1
	Upper bound	Ub ()	1
	Lower bound	Lb ()	0
	No. of iterations		20
	No. of runs		10, 20, 30, 40, 50
SVM	KernelFunction	KernelFunction	Rbf
Decision tree	Fit function	Fitctree	Fitctree (selectTrain, YTrain)

4.1.3. Evaluation metrics

The effectiveness of the bGWbPS-IDS to detect abnormal traffic is evaluated using accuracy, detection rate, false alarm rate, number of features, and processing time that are commonly used in related studies to validate the experiments (Gautam & Om, 2016; Ahmad et al., 2019; Chen, 2020).

Accuracy, detection rate, and false alarm rate are derived directly from the confusion matrix shown in Table 4. The confusion matrix contains true-positive (TP), true-negative (TN), false-positive (FN), and false-negative (FN). TP is the total number of abnormal traffic classified as abnormal traffic correctly, while TN is the total number of normal network traffic correctly classified as normal network traffic. False-positive (FP) is the total number of normal network traffic classified wrongly as abnormal network traffic, while FN is the total number of abnormal network traffic classified wrongly as normal network traffic. The following equations (Eqs (10)–(14)) are used to calculate these metrics' values, respectively.

$$Average\ \ Accuracy = \frac{\sum\left(\frac{TP+TN}{TP+TN+FP+FN}\right)}{Total_{Runs}} \tag{10}$$

Table 3
Statistics of Benchmarking Datasets.

Dataset	Category	Training set	Testing set
NSL KDD'99	Normal	13,449	2152
	DoS	9234	4344
	Prob	2289	2402
	R2L	11	2884
	U2R	209	68
	Total	25,192	11,850
UNSW-NB15	Normal	56,000	37,000
	Analysis	2000	677
	Backdoor	1746	583
	DoS	12,264	4089
	Exploits	33,393	11,132
	Fuzzers	18,184	6062
	Generic	40,000	18,871
	Reconnaissance	10,491	3496
	Shellcode	1133	378
	Worms	130	44
	Total	**175,341**	**82,332**

Table 4
Confusion Matrix.

	Abnormal	Normal
Abnormal	TP	FP
Normal	FN	TN

Table 5
Hardware and software Specifications.

Item	Specification
Central Processing Unit	Intel® Core™ i5-4590 CPU @3.30 GHz
Random Access Memory	8 GB
Hard Disk Drive	1 TB
Operating system	Microsoft Windows 10
Others	MATLAB, MS-Excel 2016

$$Average\ \ Detection\ \ rate = \frac{\sum\left(\frac{TP}{TP+FN}\right)}{Total_{Runs}} \tag{11}$$

$$Average\ \ False\ \ alarm\ \ rate = \frac{\sum\left(\frac{FP}{TN+FP}\right)}{Total_{Runs}} \tag{12}$$

The number of features is generated automatically from the simulation by calculating the average number of generated features used in each run. The processing time is measured by calculating the time intervals between the starting and the ending time of the experiment. The following equations are used to calculate the average number of features and processing time, respectively.

$$Average\ \ Number\ \ of\ \ features = \frac{Total_{Features}}{Total_{Runs}} \tag{13}$$

where $Total_{features}$ denotes the total number of features generated from all runs, and $Total_{runs}$ denotes the total number of runs in each experiment.

$$Processing\ \ time = RE - RS \tag{14}$$

where RE denotes the end-run time in millisecond, and RS denotes the start run time in millisecond.

4.1.4. Hardware and software specifications

This section presents the hardware and software specifications used to implement and test the effectiveness of bGWbPS-IDS. Table 5 summarizes the hardware specifications.

4.1.5. Testing approaches

Two main testing approaches are used to evaluate the effectiveness of the classifiers: (i) cross-validation test, and (ii) supplied set test.

By using the cross-validation test, the classifiers are first trained using a portion of the dataset before testing the rest of the dataset. As shown in Fig. 8, each dataset (NSL KDD'99 and UNSW-NB15) is split into two parts based on the most used percentages used in the literature (i.e., 80% for training, and 20% for testing).

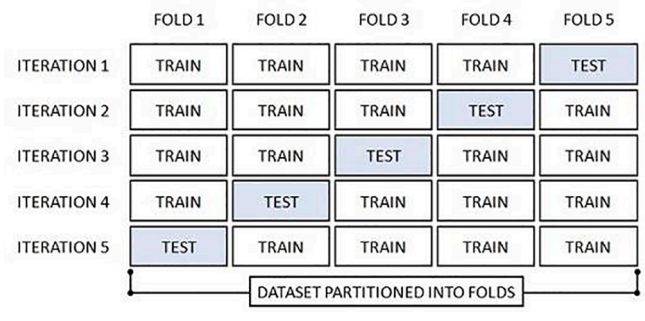


Fig. 8. Representation of Five-fold Cross-validation.

Whilst, using the second testing approach (i.e., the supplied set test approach), as shown in Fig. 9, the classifier is trained on a dataset and tested on a totally different dataset intentionally created for testing purposes. In other words, using this testing approach, the flows in the testing dataset are totally different from the flows in the training dataset.

4.2. Results and analysis

As aforementioned, the bGWbPS-IDS technique is evaluated using two benchmark datasets, NSL KDD'99 and UNSW-NB15 (Moustafa and Slay, 2016). Each dataset experiment was performed with different runs to meet the requirements of computer science's test (Devore, 2015). The bGWbPS-IDS technique's performance is compared with the existing IDS techniques, such as GWO-IDS, bGWO-IDS, bPSO-IDS, and GWOGA-IDS, to determine the effectiveness of the proposed IDS. They achieved comparable results, based on the accuracy, detection rate, false alarm rate, number of features, and processing time obtained from the previous scenarios.

Using the supplied set testing approach (i.e., training dataset is separate from testing dataset), Table 5 summarizes the results obtained from bGWbPS-IDS, GWO, bGWO, bPSO, and GWO-GA techniques using the NSL-KDD'99 dataset.

As shown in Table 4, the results reveal that the proposed IDS technique outperforms the existing IDS techniques in terms of average accuracy, average detection rate, average false alarm rate, average number of features, and average processing time, as it achieved the highest average accuracy (i.e., 64.328% and 70.956% with the SVM and decision tree, respectively) and highest average detection rate (60.268% and 68.288% with the SVM and decision tree, respectively), and lowest false alarm rate (i.e., 0.152% and 0.17% when the SVM and decision tree are used, respectively) and lowest average number of features (i.e., 16 and 18 features when used with SVM and decision tree, respectively). In addition, bGWbPS-IDS consumes the least amount of processing time among the existing IDS techniques. It takes 27715.4 s and 3396.2 s to detect all intrusions within the NSL-KDD'99 dataset when using the SVM and decision tree, respectively.

As shown in Table 6, the results reveal that the proposed IDS outperforms the existing IDS techniques in terms of average accuracy, average detection rate, average false alarm rate, the average number of features, and average processing time. It achieves the highest average accuracy (i.e., 98.82% and 99.058% when used with the SVM and decision tree, respectively). It also has the highest average detection with the SVM (98.554%) and decision tree (99.018%) and lowest false alarm rate (i.e., 0.0001% and 0.0001% when used with the SVM and decision tree, respectively) and lowest average number of features (i.e., 14.2 and 10.2 features when used with the SVM and decision tree, respectively). The bGWbPS-IDS consumes the least amount of processing time among the existing IDS techniques to detect all intrusions within the NSL-KDD'99 dataset using the SVM (24682.6 s) and decision tree (2103.6 s).

Table 7 summarizes the results obtained from bGWbPS-IDS, GWO, bGWO, bPSO, and GWO-GA techniques using the UNSW-NB15 dataset with the supplied set testing approach. The results reveal that the proposed IDS technique outperforms the existing IDS techniques in average accuracy, average detection rate, average false alarm rate, the average number of features, and average processing time as it achieves the highest average accuracy (i.e., 82.762% and 86.996% when used with the SVM and decision tree, respectively) and highest average detection rate (98.994% and 96.442% when used with the SVM and decision tree, respectively). Furthermore, the proposed technique achieves the lowest false alarm rate (i.e., 0.378% and 0.246% when used with the SVM and decision tree, respectively) and lowest average number of features (i.e., 16 and 12 features when used with the SVM and decision tree, respectively). The bGWbPS-IDS consumes the least processing time among the existing IDS techniques since it consumes 49647.6 s and 26,999 s to

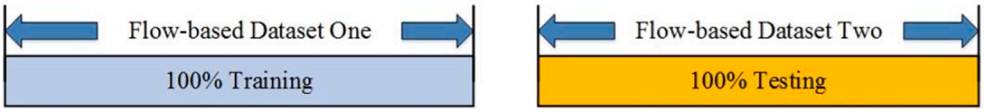


Fig. 9. Representation of Supplied Set Test Approach.

Table 6
bGWbPS-IDS vs. Exiting IDS Techniques using NSL KDD'99 dataset and Supplied Set Testing Approach.

Dataset	IDS Technique	Classifier	Avg. Accuracy	Avg. Detection Rate	Avg. False Alarm	Avg. No. of Features	Avg. Processing Time
NSL KDD'99	GWO	SVM	57.834	53.278	0.21754	21	28894.0864
		DT	64.934	60.9	0.174	22	3642.2
	bGWO	SVM	61.016	55.382	0.158	28	29385.6
		DT	70.04	66.614	0.176	25	3201.8
	bPSO	SVM	64.126	60.128	0.17	18	28405.2
		DT	70.784	68.21	0.176	21	2618.8
	GWO-GA	SVM	61.13	56.626	0.18	20	29,345
		DT	69.094	66.15	0.178	20	2909.2
	bGWbPS (Proposed)	**SVM**	**64.328**	**60.268**	**0.152**	**16**	**27715.4**
		DT	**70.956**	**68.288**	**0.17**	**18**	**3396.2**

Table 7
bGWbPS-IDS vs. Exiting IDS Techniques using NSL KDD'99 dataset Cross-Validation Testing Approach.

Dataset	IDS Technique	Classifier	Avg. Accuracy	Avg. Detection Rate	Avg. False Alarm	Avg. No. of Features	Avg. Processing Time
NSL KDD'99	GWO	SVM	98.362	97.648	0.0001	20.6	25667.2
		DT	98.47	98.376	0.0001	15.6	2176.8
	bGWO	SVM	98.404	97.666	0.0002	29.2	27024.4
		DT	99.006	98.834	0.0004	14.6	2193.4
	bPSO	SVM	98.368	98.276	0.0001	22.6	25611.8
		DT	98.996	98.918	0.0003	16.4	2216.2
	GWO-GA	SVM	98.176	97.36	0.0001	24	26665.2
		DT	98.7	98.666	0.0001	22.6	2343
	bGWbPS (Proposed)	**SVM**	**98.82**	**98.554**	**0.0001**	**14.4**	**24682.6**
		DT	**99.058**	**99.018**	**0.0001**	**10.2**	**2103.6**

detect all intrusions within the UNSW-NB15 dataset using the SVM and decision tree, respectively.

Table 8 summarizes the results obtained from bGWbPS-IDS, GWO, bGWO, bPSO, and GWO-GA techniques using the UNSW-NB15 dataset with cross-validation. The results reveal that the proposed IDS outperforms the existing IDS techniques in average accuracy, average detection rate, average false alarm rate, average number of features, and average processing time. It achieves the highest average accuracy (i.e., 94.78% and 94.482% when it used with SVM and decision tree, respectively) and highest average detection rate (98.75% and 96.416% when it used with SVM and decision tree, respectively), and lowest false alarm rate (i.e., 0.154% and 0.1% when it used with SVM and decision tree, respectively) and lowest average number of features (i.e., 12.6 and 11 features when it used with SVM and decision tree, respectively). In addition, bGWbPS-IDS consumes the least amount of processing time among the existing IDS techniques since it consumes 37060.6 s and 23906.6 s to detect all intrusions within the UNSW-NB15 dataset when used with the SVM and decision tree, respectively, as shown in Table 9.

In sum, from the findings, it can be concluded that the bGWbPS-IDS is indeed a practical method to address IDS problems. The algorithm's ability to increase the accuracy value and decrease the number of features for the detection process significantly enhanced the performance of the IDS. It was discovered that the added multi-objective function (i.e., fitness function) with the fourth grey wolves have a direct impact on the algorithm's next position selection process. Compared to other existing IDS techniques, the obtained experimental outcomes demonstrated that the proposed bGWbPS-IDS had a higher positive impact on enhancing IDS performance.

In fact, adding one more wolf (i.e., omega wolf (ω)) in bGWO provides accurate decisions and reduces decision dependency. So, the

Table 8
bGWbPS-IDS vs. Exiting IDS Techniques using UNSW-NB15 Supplied Set Testing Approach.

Dataset	IDS Technique	Classifier	Avg. Accuracy	Avg. Detection Rate	Avg. False Alarm	Avg. No. of Features	Avg. Processing Time
UNSW-NB15	GWO	SVM	81.66	98.89	0.404	19	58484.4
		DT	86.758	96.334	0.25	18	33,193
	bGWO	SVM	82.504	98.95	0.38	22	42615.8
		DT	86.928	96.966	0.25	19	24699.6
	bPSO	SVM	82.62	98.872	0.384	24	41285.2
		DT	86.914	96.376	0.248	21	24642.2
	GWO-GA	SVM	81.372	98.916	0.42	24	57314.2
		DT	86.932	96.106	0.25	21	24830.6
	bGWbPS (Proposed)	**SVM**	**82.762**	**98.994**	**0.378**	**16**	**49647.6**
		DT	**86.996**	**96.442**	**0.246**	**12**	**26,999**

Table 9
bGWbPS-IDS vs. Exiting IDS Techniques using UNSW-NB15 Cross-Validation Testing Approach.

Dataset	IDS Technique	Classifier	Avg. Accuracy	Avg. Detection Rate	Avg. False Alarm	Avg. No. of Features	Avg. Processing Time
UNSW-NB15	GWO	SVM	93.758	98.574	0.168	16	46,605
		DT	93.882	96.178	0.1	17.6	31685.2
	bGWO	SVM	94.204	98.648	0.162	15.6	39324.6
		DT	93.82	95.604	0.096	15.6	26811.6
	bPSO	SVM	94.018	98.61	0.166	21.6	37688.4
		DT	93.932	95.838	0.094	20.6	24,202
	GWO-GA	SVM	91.25	98.7	0.184	19	45,687
		DT	93.9	95.738	0.092	18.6	24307.8
	bGWbPS (Proposed)	**SVM**	**94.78**	**98.75**	**0.154**	**12.6**	**37060.6**
		DT	**94.482**	**96.416**	**0.1**	**11**	**23906.6**

bGWO is unrestricted to GWO for the FS problem. However, in a modified bGWO, the next position changed based on the four best solutions of α, β, δ, and ω using the crossover approach. The use of multi-objective function also leads to an appropriate selection of a set of features. It also evaluates the feature subset if it meets the objectives (i.e., high detection accuracy and a minimum number of features) or not, effectively. Then, the premature convergence and avoiding a local minimum trap due to diversity loss; the performance of bPSO is improved by utilizing the concepts of modified bGWO to enrich the exploration process (i.e., hybridization).

5. Conclusion and future works

bGWbPS-IDS that improve the FS process is proposed in this study. A modified change position technique was added to the bGWO to improve the performance of IDS FS process. Additionally, a multi-objective fitness function which aims to select a small subset of features with high detection accuracy was added to the bGWO. IDS FS process is further improved by hybridizing bGWO with bPSO.

Meanwhile, the proposed IDS is evaluated on two different benchmarking datasets (NSL KDD'99 and UNSW-NB15) with two testing approaches (i) supplied set testing approach and, (ii) cross-validation testing approach. The experiments were set up with specific parameters, the values of which were either taken from previous studies or determined experimentally. Furthermore, the proposed IDS performance is compared to existing IDS techniques such as GWO-IDS, bGWO-IDS, bPSO-IDS, and GWOGA-IDS in terms of detection accuracy, detection rate, false alarm rate, number of features, and processing time to determine the effectiveness of the proposed IDS.

The experimental results revealed that the proposed IDS improves the detection accuracy by approximately 0.3% to 12%, and the detection rate by 2% to 12%. In addition, it reduces false alarm rates by 4% to 43%, and reduces the number of features by approximately 31% to 75%. Last, the proposed IDS reduces processing time by approximately 14% to 22% compared to state-of-that-art approaches. The proposed IDS has also opened interesting future research directions for anomaly detection. Several future work directions including the next location decision can be further enhanced by adapting the PSO algorithm's velocity parameter. Three wolves of the original GWO may participate in the next location decision, like local-best and global-best locations, upon participating in the PSO algorithm.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix A. Data transformation of UNSW-NB15 Dataset

Symbolic data	Numeric data	Symbolic data	Numeric data
proto-Transaction		Hmp	34
3pc	1	Iatp	35
a/n	2	Ib	36
aes-sp3-d	3	Icmp	37
Any	4	Idpr	38
Argus	5	idpr-cmtp	39
Aris	6	idrp-	40
Arp	7	Ifmp	41
ax.25	8	Igmp	42
bbn-rcc	9	Igp	43
Bna	10	Il	44
br-sat-mon	11	i-nlsp	45
Cbt	12	Ip	46
Cftp	13	Ipcomp	47
Chaos	14	Ipcv	48
compaq-peer	15	Ipip	49
Cphb	16	Iplt	50
Cpnx	17	Ipnip	51
Crtp	18	Ippc	52
Crudp	19	ipv6	53
Dcn	20	ipv6-frag	54
Ddp	21	ipv6-no	55
Ddx	22	ipv6-opts	56
Dgp	23	ipv6-route	57
Egp	24	ipx-n-ip	58

(*continued on next page*)

(*continued*)

Symbolic data	Numeric data	Symbolic data	Numeric data
Eigrp	25	Irtp	59
Emcon	26	Isis	60
Encap	27	iso-ip	61
Etherip	28	iso-tp4	62
Fc	29	Kryptolan	63
Fire	30	l2tp	64
Ggp	31	Larp	65
Gmtp	32	leaf-1	66
Gre	33	leaf-2	67
Pipe	82	Xnet	130
Pnni	83	xns-idp	131
pri-enc	84	Xtp	132
Prm	85	Zero	133
Ptp	86	Xnet	130
Pup	87	xns-idp	131
Pvp	88	service	
Qnx	89	–	0
Rdp	90	Dhcp	1
Rsvp	91	Dns	2
Rtp	92	ftp	3
Rvd	93	ftp-data	4
sat-expak	94	http	5
sat-mon	95	Irc	6
Sccopmce	96	pop3	7
Scps	97	Radius	8
Sctp	98	Smtp	9
Sdrp	99	Snmp	10
secure-vmtp	100	Ssh	11
Sep	101	Ssl	12
Skip	102	stat	
Sm	103	–	0
Smp	104	Acc	1
Snp	105	Clo	2
sprite-rpc	106	Con	3
Sps	107	Eco	4
Srp	108	Fin	5
st2	109	Int	6
Stp	110	No	7
sun-nd	111	Par	8
Swipe	112	Req	9
Tcf	113	Rst	10
Tcp	114	Urn	11
Tlsp	115		
tp++	116		
trunk-1	117		
trunk-2	118		
Ttp	119		
Udp	120		
Unas	121		
Uti	122		
vines	123		
Visa	124		
Vmtp	125		
vrrp	126		
wb-expak	127		
wb-mon	128		
wsn	129		

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